IIMS26 Sensor Processing Report Template

June 2026

1 Introduction

- provide context
  - what was measured?
  - what sensor was used?
  - how was the sensor deployed?
  - where did the measurement take place?
  - is additional information available (and potentially used)?
- what is your objective for the sensor processing?

2 Methodology

- show the concept of your processing pipeline
- why did you decide to use the approach you chose?
- how did you implement your processing pipeline?
- define your metrics for analysis:
  - are you assessing the processing results qualitatively or quantitatively? Why are you choosing one or the other or both?
  - if you use quantitative metrics what are they and how are they defined? (e.g. root mean square error between processed data and ground truth)

3 Results and Discussion

- show your results
  - figures need to be self-explanatory. This means you need figure captions that provide enough information to understand the figure without reading the main text.

  - axes of plots need to be labeled (minimum font size of 8 pts) with a meaningful name of the variable (or an explanation in the caption if a single letter is used) including units (if the variable is expressed in a physical unit)
  - every used figure needs to be meaningful for the analysis and discussion in the main text and therefore every used figure should be referenced in the text.

- analyze your results i.e. what are the important aspects to see in your figures?
- discuss your results
  - what do the results mean in the context of your objective?
  - what are potential uncertainties and limitations in your data processing?

4 Conclusion and Further Considerations

- what is the outcome of your processing?
- what are things one could consider with additional time, additional context or additional measurements?
- add additional thoughts that you have on your data processing if you have any

5 Bibliography (if necessary)

Every visual and scientific concept or algorithm that has not been produced by yourself needs to be properly referenced in text with the full reference appearing here in the bibliography. Exceptions are concepts that are sufficiently standard and widespread (i.e. you do not need to reference a Fast Fourier Transform or the calculation of a mean).

6 Contribution of each Group Member

Please indicate who in your group did contribute to which aspect of the project. It is fine for more than one person to work on the same aspect. However, everyone should contribute something to the processing itself (i.e. it is not enough to just write the report or make the presentation).